MALLA REDDY ENGINEERING COLLEGE FOR WOMEN

B.Tech. IV Year II Sem		L	T	P	C
	2205PE09	3	0	0	3

Cloud Security
(Professional Elective VI)

Course Description:

- This course will provide a foundational understanding of what is required to secure a cloud ecosystem, regardless of the vendor. The concepts and principles discussed will help bridge the gaps between traditional and cloud security architectures while accounting for the shifting thought patterns involving enterprise risk management.

Course Educational Objectives:

- Understand fundamental cloud computing concepts and deployment models.
- Understand the foundational security practices that are required to secure modern cloud computing infrastructures.
- Understand the security risks involved in the cloud environment.
- Learn how attempt is made to resolve the challenges in the cloud environment.
- Understand the Cloud Security Architecture and Design patterns.

Unit-1:
Introduction to Cloud: Cloud Delivery Models, Cloud Deployment Models, Cloud Computing Software Security Fundamentals: Cloud Information Security, Objectives, Cloud Security Service

Unit-II:
Cloud Security Principles in all steps: Cloud Security Design Principles, Secure Cloud, Software Requirements, Approaches to Cloud Software Requirements Engineering,Cloud Security Policy Implementation and Decomposition, Secure Cloud Software Testing, Cloud Penetration Testing

Unit-III:
Cloud Computing Risk Issues: The CIA Triad, Privacy and Compliance Risks, Common Threats and Vulnerabilities, Cloud Access Control Issues, Cloud Service Provider Risks.

Unit-IV:
Cloud Computing Security Challenges: Security Policy Implementation, Policy Types, Computer Security Incident Response Team (CSIRT), VM Security Recommendations.

Unit-V:
Cloud Computing Security Architecture and Design patterns, Architectural Considerations, Trusted Cloud Computing, Secure Execution Environments and

Communications, Identity Management and Access Control, Autonomic Security, Introduction to Design Patterns,Security Patterns for Cloud Computing

Textbooks:

1. Ronald L. Krutz Russell Dean Vines , Cloud Security - A Comprehensive Guide to Secure Cloud Computing Published by Wiley Publishing, Inc.
2. Securing The Cloud: Cloud Computing Security Techniques and Tactics by Vic (J.R.) Winkler (Syngress/Elsevier)

References:

1. Chris Dotson, Practical Cloud Security: A Guide for Security Design and Deployment, Published by O'Reilly, 1st Edition.
2. John Vacca, Cloud Computing Security: Foundations and Challenges, 2nd Edition CRC Press.
3. Cloud Computing Design Patterns by Thomas Erl (Prentice Hall)